Higher National Diploma in Engineering (Electrical and Electronics)

First Year, First Semester Examination — 2019

Electrical Principles B

Question 1

1. (i). Briefly explain the majority and minority carriers of p and n type semiconductors.

1. (ii). Explain the operation of a semiconductor diode during forward and reverse biased conditions.

1. (iii). Draw the waveform of the output voltage V_0 for Figure Q1a by considering that the diode is ideal, $V_{in} = V_m \sin \omega t$ and $V < V_m$.

1. (iv). A 10 V Zener diode is used as a voltage regulator & load resistance is $3\ \mathrm{k}\Omega$ as shown in Figure Q1b. **Find the followings.**

- a). The load voltage
- b). The Zener current
- c). The power dissipated in the Zener diode
- d). Explain the operation of the circuit as a voltage regulator with different loads for the same input.

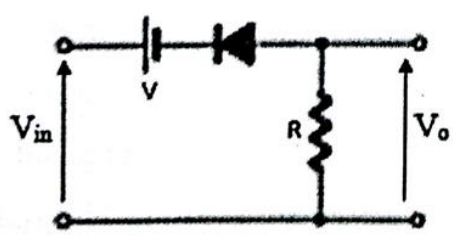

Figure Q1a

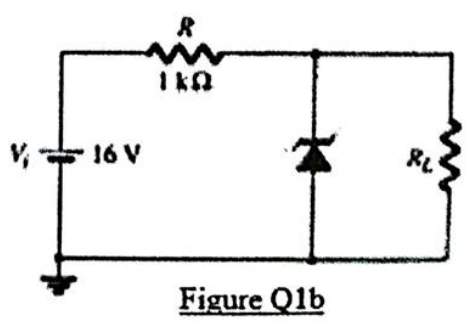

Figure Q1b

Question 2

2. (i). Explain importance of the peak inverse voltage of the diode.

2. (ii). The bridge rectifier shown in Figure Q2a has four ideal diodes and connected $R = 2\ \mathrm{k}\Omega$ as load resistance. The supply voltage is $10 \sin 300t$. Calculate,

- **a). the output dc voltage**
- b). Efficiency of the circuit
- c). PIV of the diode

2. (iii). The voltage regulator circuit is modified as shown in Figure Q2b.

- a). Sketch the waveform of the output voltage (V_o) for the same input voltage.
- b). Calculate the output DC voltage and PIV

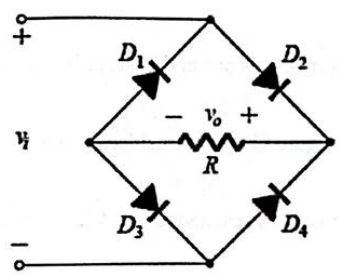

Figure Q2a

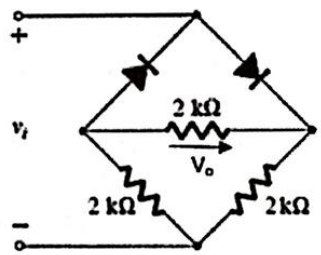

Figure Q2b

Question 3

3. (i). Explain why V_{CE} is usually chosen around $V_{CC}/2$ in amplifiers.

3. (ii). CE amplifier circuit is shown in Figure Q3 & Si transistor is used with $\beta = 100$.

- a). Find DC operating point of the amplifier.
- b). Draw ac equivalent circuit considering very high input resistance of the transistor.
- c). Calculate voltage gain, current gain & input resistance.

3. (iii).

Draw ac equivalent circuit by removing C_2 capacitor in Figure Q3 & calculate voltage gain.

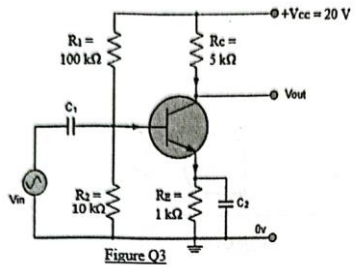

Figure Q3

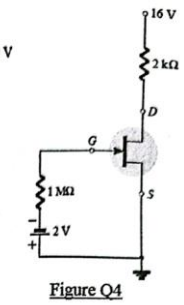

Figure Q4

Question 4

4. (i). Explain briefly the necessity of negative V_{GS} voltage to the operation of JFET.

4. (ii). a. Define trans conductance (g_m) for JFET and derive expression for g_m with the terms I_D, V_P, and I_{DSS}.

- b. A JFET has parameters $I_{DSS} = 5\ \mathrm{mA}$, $V_P = -2\ \mathrm{V}$. Find the value of drain current and g_m at $V_{GS} = -1.5\ \mathrm{V}$.

4. (iii).

A fixed biased JFET amplifier having parameters $I_{DSS} = 10\ \mathrm{mA}$ & $V_P = -8\ \mathrm{V}$ is shown in Figure Q4. **Determine V_{GS}, I_D and V_{DS}.**

Question 5

5. (i). Explain the difference between combinational logic circuit and sequential logic circuit.

5. (ii). Convert the following numbers with the indicated bases to binary.

- a. 36.375_{10}
- b. 374.2_8
- c. $17\mathrm{E.F}_{16}$

5. (iii). Find the values of followings using 8 bit binary form of 2's complement method.

- a. $37 - 18$
- b. $-80 - 32$

5. (iv). A logic circuit is shown in Figure Q5 has four inputs A, B, C, D and output is Y.

- a. Draw the truth table
- b. Write a Boolean expression for output
- c. Simplify the expression

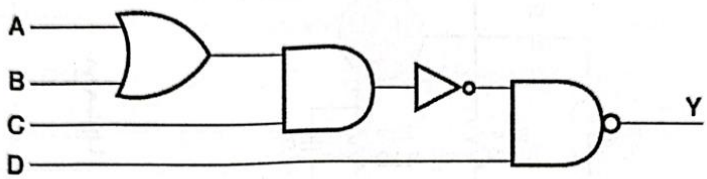

Figure Q5

Question 6

6. (i). State de Morgan's theorem for two logic variables X and Y.

6. (ii). Draw a logic diagram for function $F = (A' + B')(C + D)$ using

- **a. only NOR gates**
- **b. only NAND gates**

6. (iii). Use a Karnaugh map to simplify the followings.

- a. $X = (AB + A'C + ABC)$
- b. $Y = (0, 1, 2, 5, 6, 7, 9, 11, 12, 14)$ (This appears to be a list of minterms for a four-variable function)

6. (iv). Write the short notes on followings.

- a. Logic gates
- b. Karnaugh map
- c. Redundant group